

FYS-STK3155/4155 SEP 7

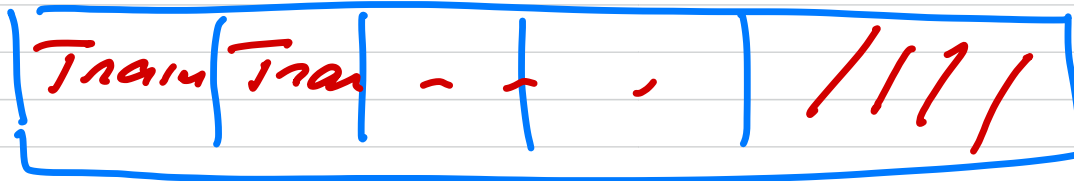


FYS-STK3155/4155 SEP 7

k-fold of CV

$$k = 5$$

Test



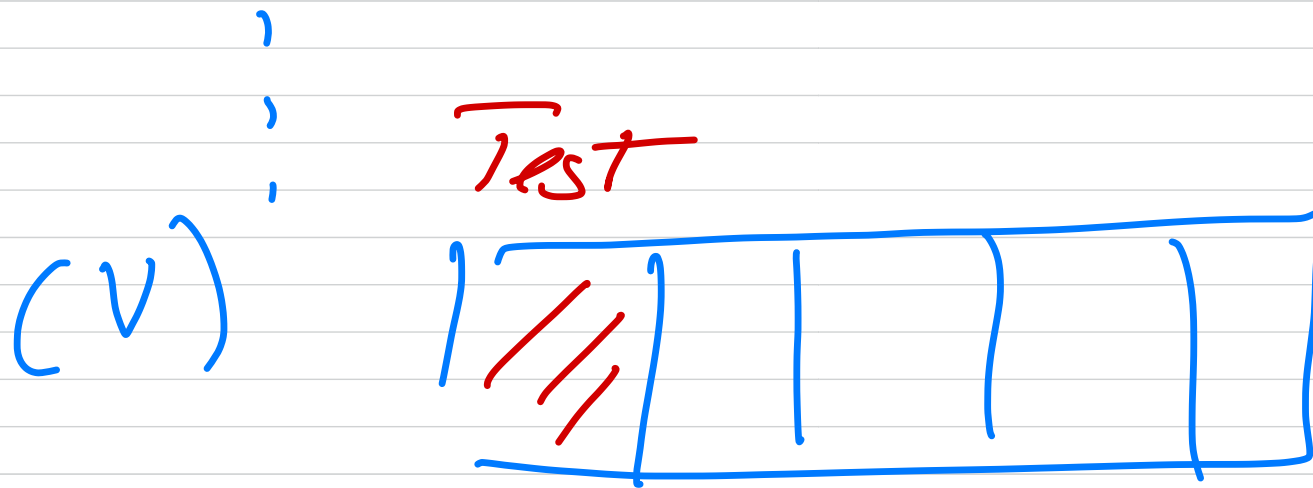
$$MSE(i) = \frac{1}{N_{train} - 1} \sum_{n=1}^{N_{train}-1} (y_n - \hat{y}_n)^2 \quad (i)$$

$$MSE_{test}(i)$$

Test



$$MSE_{test}(i)$$



$MSE_{Test}(v)$

$$MSE_{Test} = \frac{1}{5} \sum_{i=1}^5 MSE_{Test}(v_i)$$

Optimization

OLS gradient

$$\vec{\nabla}_{\theta} C(\theta) = -\frac{2}{n} X^T (y - X\theta)$$

$$\nabla_{\theta}^2 C(\theta) = \frac{2}{n} X^T X \quad (\text{Hessian})$$

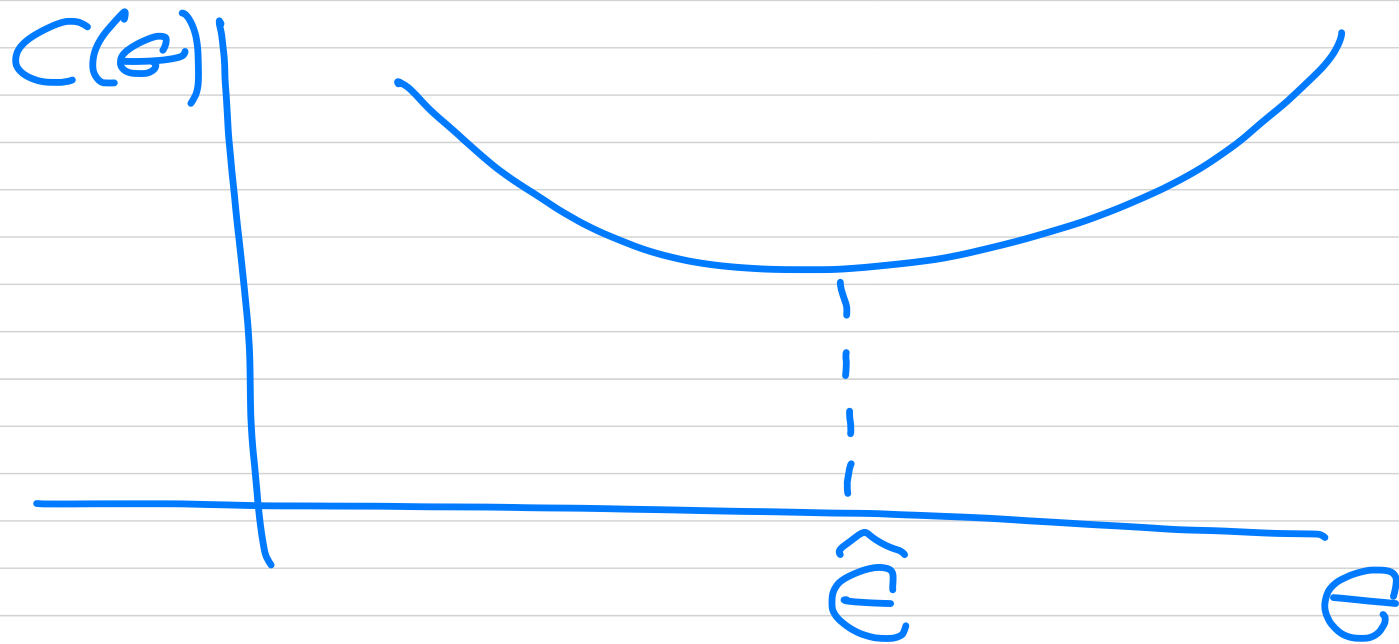
semi-positive definite $\Rightarrow$
global minimum.

Ridge Regression

$$\vec{\nabla}_{\theta} C(\theta) = -\frac{2}{n} X^T (y - X\theta) + 2\lambda \theta$$

$$\nabla_{\theta}^2 C(\theta) = 2 \left(X^T X / n + \lambda \right) \quad \lambda \geq 0$$

Newton's Method



$$\frac{dC}{d\theta} = g$$

Taylor expansion around
 $\hat{\theta} = \hat{\theta} - \theta^{(n)}$

$$\begin{aligned}
 C(\hat{e}) = & C(e^{(n)}) + \\
 & g(e^{(n)}) (\hat{e} - e^{(n)}) + \\
 & \frac{1}{2} \left. \frac{d^2 C}{de^2} \right|_{e=e^{(n)}} (\hat{e} - e^{(n)})^2
 \end{aligned}$$

+ ~ ~ ~

Truncate at $\frac{d^2 C}{de^2}$

$$\begin{aligned}
 C(\hat{e}) &\approx C(e^{(n)}) + \underbrace{g(e^{(n)})}_{g^{(n)}} (\hat{e} - e^{(n)}) \\
 &+ \frac{1}{2} \underbrace{\left. \frac{d^2 C}{d e^{(2)}} \right|_{e=e^{(n)}}}_{H^{(n)}} \underbrace{(\hat{e} - e^{(n)})^2}_{b^{(n)}}
 \end{aligned}$$

$$\begin{aligned}
 C(\hat{e}) &= C(e^{(n+1)}) = \\
 &C(e^{(n)}) + g^{(n)} b^{(n)} + \frac{1}{2} H^{(n)} (b^{(n)})^2
 \end{aligned}$$

optimize wrt $t^{(n)}$

$$\frac{dc}{dt^{(n)}} = 0 = g^{(n)} + t^{(n)} H^{(n)}$$

$$t^{(n)} = e^{(n+1)} - e^{(n)}$$

$$= - [H^{(n)}]^{-1} g^{(n)}$$

$$\Rightarrow e^{(n+1)} = e^{(n)} - (H^{(n)})^{-1} g^{(n)}$$

$$\frac{dC}{d\mathbf{e}} \rightarrow \vec{\nabla}_{\mathbf{e}} C(\mathbf{e}) \big|_{\mathbf{e} = \mathbf{e}^{(n)}} \\ = \vec{\nabla}_{\mathbf{e}} C(\mathbf{e}^{(n)})$$

Hessian matrix with p -features

$$H(\mathbf{e}^{(n)}) = \begin{bmatrix} \frac{\partial^2 C}{\partial \mathbf{e}_0^2} & \frac{\partial^2 C}{\partial \mathbf{e}_0 \partial \mathbf{e}_1} & \dots & \frac{\partial^2 C}{\partial \mathbf{e}_0 \partial \mathbf{e}_{p-1}} \\ \vdots & & & \\ \frac{\partial^2 C}{\partial \mathbf{e}_{p-1} \partial \mathbf{e}_0} & \dots & \dots & \frac{\partial^2 C}{\partial \mathbf{e}_{p-1}^2} \end{bmatrix}$$

Note that for OLS (and Ridge)
no ϵ -dependence in
 H

$$\text{OLS : } D_{\epsilon}^2(\epsilon) = \frac{2}{n} X^T X$$

$\nearrow$
no ϵ -dependence

$\alpha^{(n)}$

$$\epsilon^{(n+1)} = \epsilon^{(n)} - \left[H(\epsilon^{(n)}) \right]^{-1} D_{\epsilon}(\epsilon^{(n)})$$

$$E^{(n+1)} = E^{(n)} - \gamma^{(n)} \nabla_E C(E^{(n)})$$

$$\gamma^{(n)} = \boxed{\gamma \text{ constant.}}$$

Learning rate

Plain gradient descent,

algorithm:

guess for γ ; fix # of iterations

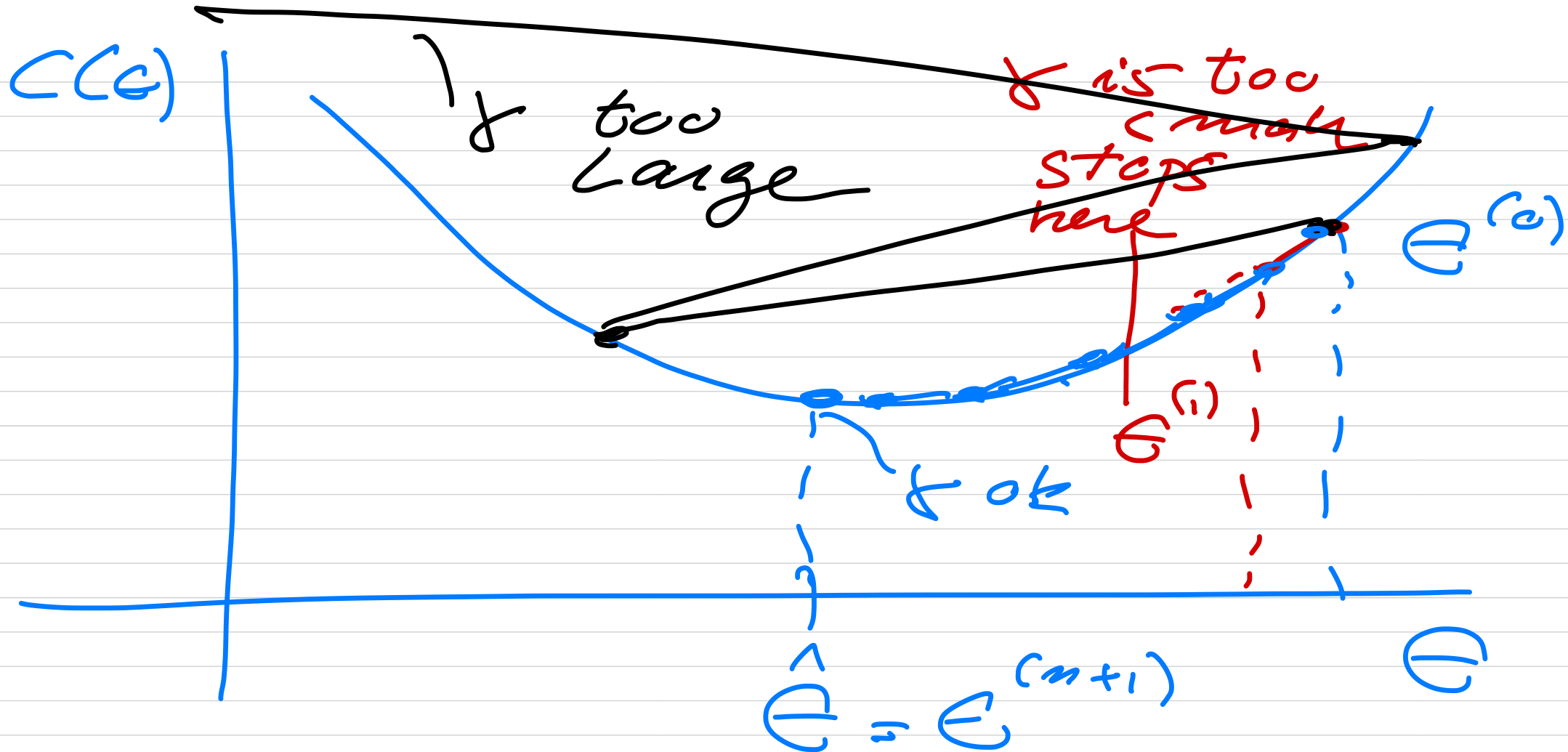
while (iterations less than

$$\max \text{ OR } |E^{(n+1)} - E^{(n)}| \leq \epsilon$$

calculate gradient

$$\text{update } E^{(n+1)} = E^{(n)} - \gamma \cdot \text{grad}$$

end while



γ : { plain GD, const γ
momentum $\gamma^{(n)}$
ADAM $\gamma^{(n)}$
RMSPROP $\gamma^{(n)}$
Adagrad $\gamma^{(n)}$
:
:

$$\begin{aligned}
C(E^{(n+1)}) &= C(E^{(n)}) \\
&+ g^{T(n)} \left(\underbrace{E^{(n)} - \gamma^{(n)} g^{(n)}}_{E^{(n+1)} - E^{(n)}} \right) \\
&+ \frac{1}{2} \left(E^{(n)} - \gamma^{(n)} g^{(n)} \right)^T H^{(n)} \left(E^{(n)} - \gamma^{(n)} g^{(n)} \right)
\end{aligned}$$

optimal γ ?

$$\frac{dC}{d\gamma} = 0 = -g^{T(n)} g^{(n)} + \gamma g^{T(n)} H^{(n)} g^{(n)}$$

$\Rightarrow$

$$\gamma^{(n)} = \frac{g^{T(n)} g^{(n)}}{g^{T(n)} H^{(n)} g^{(n)}}$$

(steepest descent)

$$H^{(n)} g^{(n)} = \sum g^{(n)}$$

$$\gamma^{(n)} = \frac{1}{\sum^{(n)}}$$

in general

$$\gamma < \frac{2}{\epsilon_{\max}}$$

where $\epsilon_{\max}$ = largest
eigenvalue of H

OLS and Ridge with small
dimensions of H , we
can easily diagonalize.

GD with momentum

$$e^{(n+1)} = e^{(n)} - \delta g(e^{(n)})$$

$$+ \delta [e^{(n)} - e^{(n-1)}]$$

δ = momentum param

$$\delta \in [0, 1]$$